

CN503 contact alignment - trial 01

Dreamcast VA1 | owner measurements, 9 October 2026

Print at actual size / 100%. Disable fit-to-page. Check BOTH 20 mm scale bars.

Alignment guide only. Electrical pin numbering and a manufacturing footprint are not shown.



Working geometry

25 contacts per row; 1.00 mm pitch

5.03 mm BETWEEN TRIAL ROW CENTERS

Approx. 0.44 mm contact width along row

24.00 mm center span (calculated)

24.44 mm outside span (calculated)

Black ticks show contact centers and width; tick length is symbolic, not measured tail length.

Dashed rectangle: 38.67 x 6.56 mm maximum housing envelope only, not a cutout.

The narrower-section width was reported as the approximate 5.03 mm row separation.

This drawing uses that value AS center spacing for a fit check; verify alignment of both rows.

How to check

1. Verify that both scale bars below measure 20 mm on the printed sheet.
2. Compare one trial row with all 25 contacts, then compare the other row.
3. Check that both rows align together. Keep the paper clear of the connector housing.

If the housing prevents a flat overlay, compare separate row strips and their marked spacing.

Working owner data

Pin run: 24.39 mm; contact width: about 0.44 mm; BIOS-to-CN503 gap: about 12 mm.

CN503 tails: essentially at motherboard surface. BIOS body: about 2 mm above it.

Flex pad overlap, exposed tail length, coverlay, solder access and tolerances remain to be designed.


20 mm - horizontal


20 mm - vertical

K-UI G1 bridge proposal | no pin orientation or connector A/B row assignment is implied.